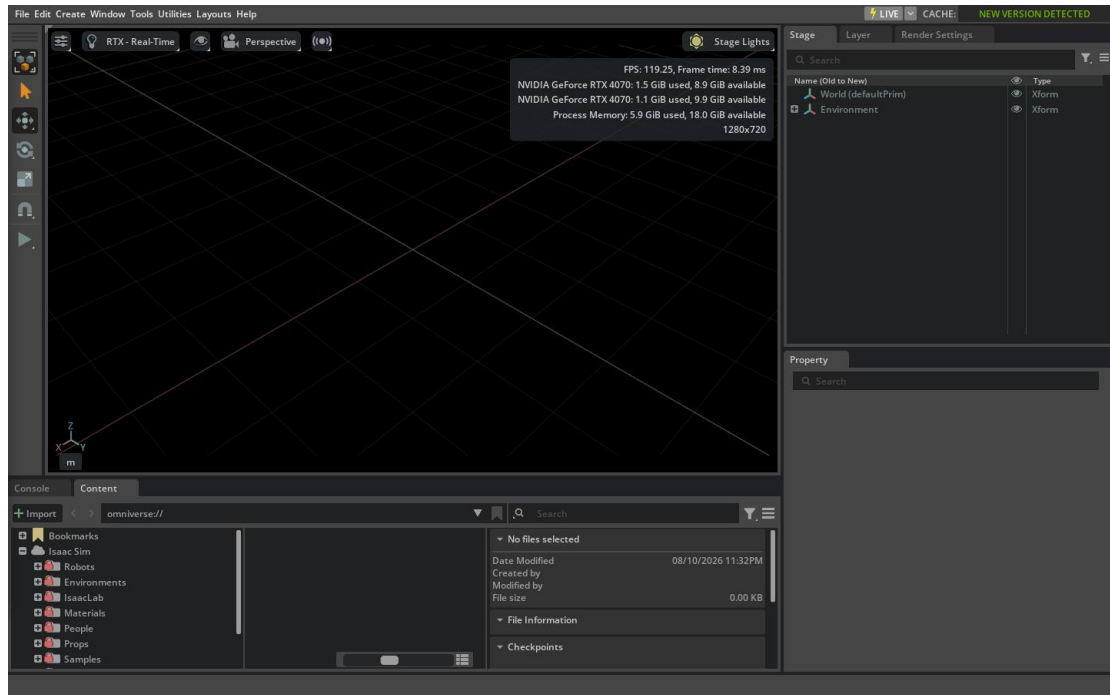


Homework1

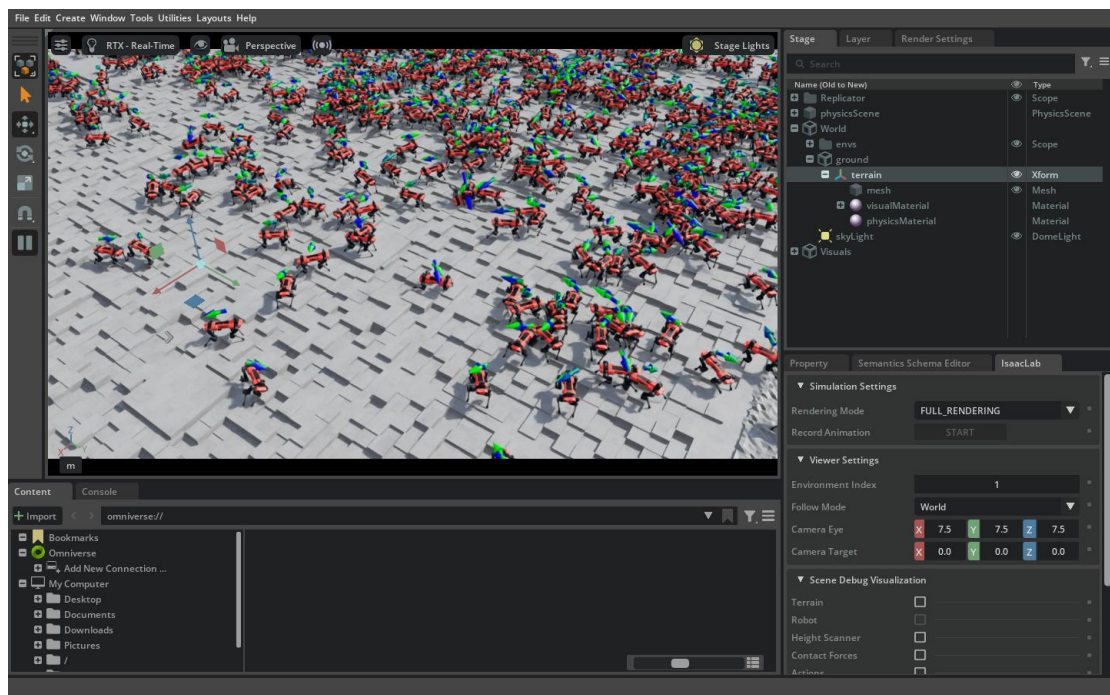
1 环境搭建结果

至少给出以下内容中的 3 项：

- Isaac Sim 成功启动截图



- Isaac Lab 场景成功启动截图



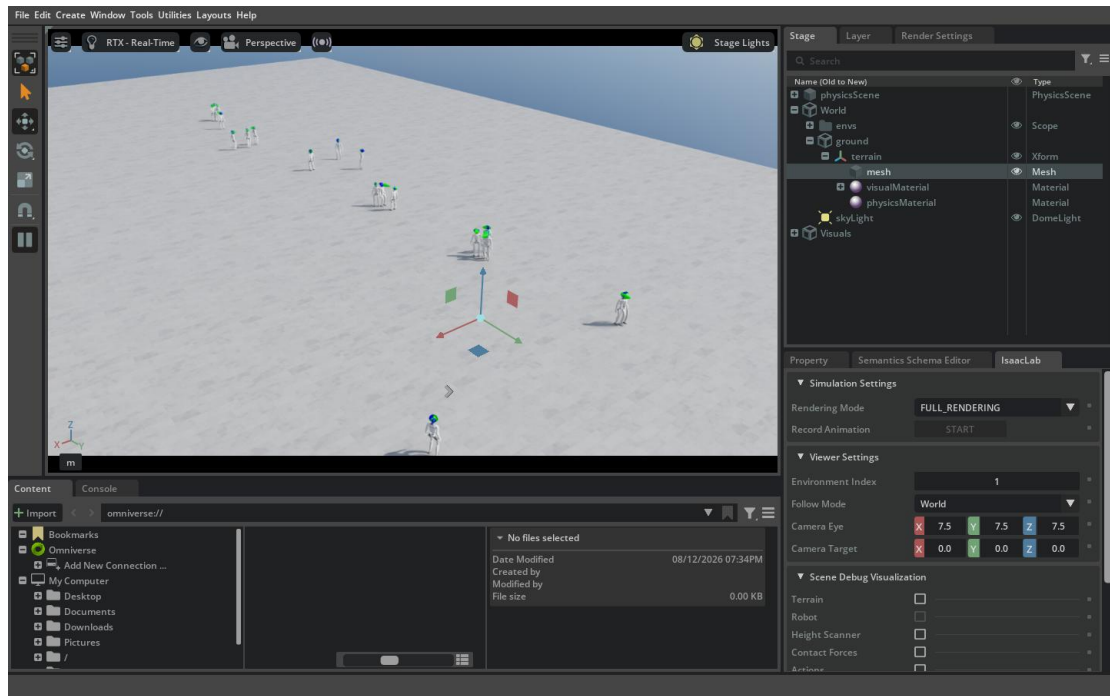
•python scripts/list_envs.py

```
star@star-A: ~/projects/unitree_rl_lab
(base) star@star-A:~/projects$ cd unitree_rl_lab/
(base) star@star-A:~/projects/unitree_rl_lab$ conda activate env_isaacLab
(env_isaacLab) star@star-A:~/projects/unitree_rl_lab$ python scripts/list_envs.py
-----
Available Environments in Unitree RL Lab
-----
S. No. | Task Name | Entry Point | Config
-----|-----|-----|-----
1 | Unitree-G1-29dof-Velocity | isaacLab.envs:ManagerBasedREnv | locomotion.robots.g1.29dof.velocity_env_cfg:RobotEnvCfg
2 | Unitree-Go2-Velocity | isaacLab.envs:ManagerBasedREnv | locomotion.robots.go2.velocity_env_cfg:RobotEnvCfg
3 | Unitree-H1-Velocity | isaacLab.envs:ManagerBasedREnv | locomotion.robots.h1.velocity_env_cfg:RobotEnvCfg
4 | Unitree-G1-29dof-Mimic-Dance-102 | isaacLab.envs:ManagerBasedREnv | mimic.robots.g1_29dof.dance_102.tracking_env_cfg:RobotEnvCfg
5 | Unitree-G1-29dof-Mimic-Gangnam-Style | isaacLab.envs:ManagerBasedREnv | mimic.robots.g1_29dof.gangnam_style.tracking_env_cfg:RobotEnvCfg
-----
(env_isaacLab) star@star-A:~/projects/unitree_rl_lab$
```

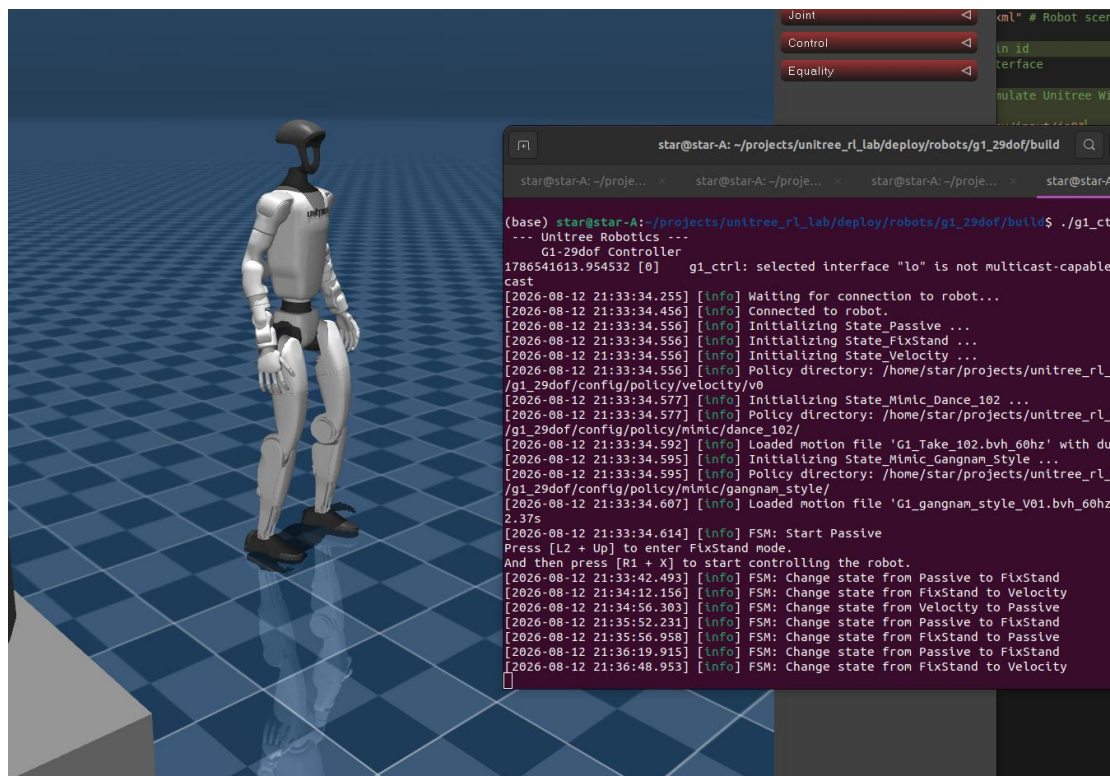
•train.py 启动训练日志截图

```
star@star-A: ~/projects/unitree_rl_lab
Time elapsed: 00:00:09
ETA: 09:53:51
#####
Learning iteration 4/50000
Computation: 45064 steps/s (collection: 2.021s, learning 0.161s)
Mean action noise std: 0.98
Mean value_function loss: 0.1610
Mean surrogate loss: -0.0132
Mean entropy loss: 40.6845
Mean reward: -4.04
Mean episode length: 42.57
Episode_Reward/track_lin_vel_xy: 0.0241
Episode_Reward/track_ang_vel_z: 0.0022
Episode_Reward/alive: 0.0071
Episode_Reward/base_linear_velocity: -0.0043
Episode_Reward/base_angular_velocity: -0.0287
Episode_Reward/joint_vel: -0.0383
Episode_Reward/joint_acc: -0.0163
Episode_Reward/action_rate: -0.1356
Episode_Reward/dof_pos_limits: -0.0010
Episode_Reward/energy: -0.0009
Episode_Reward/joint_deviation_arms: -0.0080
Episode_Reward/joint_deviation_waists: -0.0166
Episode_Reward/joint_deviation_legs: -0.0210
Episode_Reward/flat_orientation_l2: -0.0229
Episode_Reward/base_height: -0.0023
Episode_Reward/gait: 0.0115
Episode_Reward/feet_slide: -0.0090
Episode_Reward/feet_clearance: 0.0450
Episode_Reward/undesired_contacts: -0.0104
Curriculum/terrain_levels: 2.9901
Curriculum/lin_vel_cmd_levels: 0.1000
Metrics/base_velocity/error_vel_xy: 0.0482
Metrics/base_velocity/error_vel_yaw: 0.2965
Episode_Termination/time_out: 0.0004
Episode_Termination/base_height: 0.0000
Episode_Termination/bad_orientation: 0.9995
-----
Total timesteps: 491520
Iteration time: 2.18s
Time elapsed: 00:00:11
```

• play.py 运行截图



• Sim2Sim 实验过程截图



2 项目理解部分

请简要说明：

1. Unitree-G1-29dof-Velocity 的任务入口在哪个文件

unitree_rl_lab/source/unitree_rl_lab/unitree_rl_lab/tasks/locomotion/robots/g1/29dof/__init__.py

通过 gym.register 注册任务 ID，并且绑定了 RobotEnvCfg、RobotPlayEnvCfg、BasePPORunnerCfg 的路径

2. 环境配置在哪个文件

unitree_rl_lab/source/unitree_rl_lab/unitree_rl_lab/tasks/locomotion/robots/g1/29dof/velocity_env_cfg.py

3. PPO 配置在哪个文件

unitree_rl_lab/source/unitree_rl_lab/unitree_rl_lab/tasks/locomotion/agents/rsl_rl_ppo_cfg.py

4. 训练入口在哪个文件

unitree_rl_lab/scripts/rsl_rl/train.py

5. 默认训练得到的 checkpoint 大致保存在哪类目录

log_root_path = os.path.join("logs", "rsl_rl", agent_cfg.experiment_name)
对于空的 experiment_name 会保存在
unitree_rl_lab/logs/Unitree-G1-29dof-Velocity/rsl_rl/**/*.pt

3 简短分析

请用自己的话回答：

1. 从 --task 到实际环境创建的大致流程是什么

- 在 train 里面导入环境包 from list_envs import import_packages
- 在__init__里面注册任务 gym.register
- 回到 train 里面创建任务 env = gym.make(args_cli.task, cfg=env_cfg, render_mode="rgb_array" if args_cli.video else None)
- 把 IsaacLab ManagerBasedRLEnvCfg 对接到 RSL-RL 库的环境 env = RslRlVecEnvWrapper(env, clip_actions=agent_cfg.clip_actions)
- 创建训练器 runner = OnPolicyRunner(env, agent_cfg.to_dict(), log_dir=log_dir, device=agent_cfg.device)
- runner 读取 PPO 配置 RslRlOnPolicyRunnerCfg
- 开始训练 runner.learn

2. 训练配置与播放配置为什么要分开

训练环境 RobotEnvCfg 使用 scene: RobotSceneCfg = RobotSceneCfg(num_envs=4096,

env_spacing=2.5), 播放环境 RobotPlayEnvCfg 使用 self.scene.num_envs = 32 重点在于展示效果

3. play.py 为什么能够自动找到训练得到的 checkpoint

```
resume_path = get_checkpoint_path(log_root_path, agent_cfg.load_run,
agent_cfg.load_checkpoint)
```

```
from isaacsim_tasks.utils import get_checkpoint_path
```

定义在 IsaacLab 里面 IsaacLab/source/isaacsim_tasks/isaacsim_tasks/utils/parse_cfg.py
对 checkpoint 排序, 先找到最后一个 run : run_path = os.path.join(runs[-1], *other_dirs),
然后找到最后一个 checkpoint : checkpoint_file = model_checkpoints[-1]